

Evidence-Grounded Auditing of Identification Assumptions in Climate-Policy Causal Evaluations

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Abstract

Difference-in-differences (DID) studies are widely used to evaluate climate policy, but assessing the evidence supporting their identification assumptions remains challenging. We introduce ARGUS, a structured language-model pipeline that audits reported evidence against an eleven-dimension assumption–implication–evidence rubric and abstains when relevant evidence cannot be retrieved. We evaluate ARGUS using injected flaws, economics papers, and a small expert-annotated pilot. On the 11-flaw benchmark, ARGUS detects 73% of planted flaws, compared with 18% for a keyword-based pipeline. On real papers, ARGUS abstains on about 40% of dimensions for lack of retrievable evidence and, where it answers, is more severe than the experts on 28 of 33 labels. A simple post-hoc calibration rule reduces over-severity on the same pilot used to develop it. ARGUS provides evidence-linked risk reports that localize potential weaknesses for expert review, without adjudicating causal claims. Code and data: <https://github.com/yonghongzhang-io/ARGUS>.

1 Introduction

Difference-in-differences (DID) is widely used to evaluate climate and environmental policy, including carbon pricing, emissions-trading schemes, and emission standards. The credibility of these estimates depends on identification assumptions such as parallel trends, no anticipation, appropriate treatment of staggered adoption, and limited interference between units, each of which a large body of methodological work has shown to be fragile (Bertrand et al., 2004; Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; de Chaisemartin and D’Haultfœuille, 2020; Roth et al., 2023). These assumptions are often difficult to assess in climate settings, where policies overlap, implemen-

tation is staggered, spillovers cross administrative boundaries, and emissions outcomes are imperfectly measured (Appendix A). Because environmental policy increasingly rests on this empirical evidence, causal conclusions built on weakly supported assumptions carry real costs for climate governance.

Yet applied papers often assert these assumptions rather than audit whether the reported evidence supports them, one facet of a broader reliability gap in empirical research (Ioannidis, 2005; Silberzahn et al., 2018; Brodeur et al., 2020). A paper may assert parallel trends over an event-study figure, leaving the reader to judge whether the design, pre-period estimates, inference, and policy timing justify the claim. Performing this assessment manually is slow, expert-intensive, and difficult to standardize.

We argue that identification credibility is auditable even when the true causal effect is not, building on the research-design tradition in applied econometrics (Angrist and Pischke, 2010; Athey and Imbens, 2017; Rambachan and Roth, 2023). Because the counterfactual is unobserved, ARGUS does not ask whether an estimated effect is true. Instead, it asks whether each identifying assumption is adequately supported by the evidence reported in the paper, and where that support is weakest. ARGUS (after Argus Panoptes, the hundred-eyed watchman) represents a DID study through an 11-dimension *assumption–implication–evidence* rubric and uses a bounded, retrieval-gated pipeline to retrieve relevant evidence, assess its adequacy, and produce an evidence-grounded risk report for expert review (Figure 1). Model agency is confined to scoped retrieval and assessment tasks behind a fixed, deterministic control flow, unlike free-roaming agents whose control flow is model-decided (Yao et al., 2023); ARGUS flags potential weaknesses rather than adjudicating causal truth.

Evaluating such an auditor is difficult because the true policy effect is unobserved. We therefore use *flaw injection* as local ground truth: starting from a supported study fixture, we introduce one known identification flaw and test whether ARGUS detects it, remains quiet on the clean version, and localizes the affected dimension. We complement this controlled evaluation with deployment on 26 corpus papers tagged DID and a small expert-labelled diagnostic pilot used to identify calibration and applicability failures.

Contributions. Our central contribution is to make identification credibility auditable without access to the causal effect:

1. an *operationalization*: a machine-auditable *assumption–implication–evidence* rubric and a bounded, evidence-grounded pipeline with explicit abstention (§3);
2. an *evaluation methodology*: planted identification errors as local ground truth, with compute-graded and oracle ablations that attribute detection to decomposition and every synthetic omission miss to the retrieval gate (§5);
3. a *diagnosis*: on real papers, grounding limits *coverage* while over-severity limits *agreement*, and a deterministic rule derived on the pilot reduces the latter in-sample.

2 Related Work

DID identification and its threats. A large econometrics literature formalizes when DID estimators fail under staggered timing and heterogeneous effects: two-way fixed effects’ use of already-treated units as controls (Goodman-Bacon, 2021), negative weights (de Chaisemartin and D’Haultfœuille, 2020), robust estimators (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021), assumption and pre-trend surveys (Roth et al., 2023), and fragile inference under serial correlation (Bertrand et al., 2004). ARGUS adds no estimator: it *operationalizes* this literature as an auditable rubric, turning each known threat into a dimension whose supporting evidence can be checked in a given paper (§3).

LLM causal reasoning and benchmarks. A parallel line asks whether LLMs reason about causality. Benchmarks (CLadder (Jin et al.,

2023), Corr2Cause (Jin et al., 2024), CausalBN-Bench (Zhou et al., 2026)) and analyses (Kıcıman et al., 2024; Yang et al., 2024; Zečević et al., 2023) report mixed evidence: strong performance on some causal tasks, but models often reproduce causal *language* without reliable causal *inference*; a few move toward applied settings and toward separating *identification* from *estimation* (Lee et al., 2025; Sawarni et al., 2026). ARGUS asks the inverse question: are a *human* paper’s identification assumptions adequately evidenced? A causal verdict is out of scope by design.

Responsible AI for empirical research. AI is increasingly applied to research reliability: reproducibility checking, statistical-reporting validation, and AI-assisted review, motivated by wide analyst-driven result spreads (Silberzahn et al., 2018; Botvinik-Nezer et al., 2020) and specification searching in causal economics (Brodeur et al., 2020). LLMs are also used directly as evaluators and reviewers, judging model outputs (Zheng et al., 2023), giving feedback on research papers (Liang et al., 2024), and reviewing inside automated-science pipelines (Lu et al., 2024); these evaluators already produce ratings, pairwise preferences, and structured reviews. ARGUS applies an assumption-specific rubric to DID studies, links each dimension to in-paper evidence, and is measured against injected ground truth. ARGUS differs in target and posture: it audits *identification credibility* specifically (the part of an argument that licenses a causal claim) and issues no verdicts, producing evidence-grounded, human-reviewable reports that localize risk for the expert’s final call. Closest to our setting, CausalVerify (Zhang et al., 2026a) grounds evaluation in *execution*: it runs a model’s generated estimation code against synthetic data and checks the recovered effect. That benchmark does not audit identification assumptions, which is the layer ARGUS takes up; we reuse its released corpus of economics papers and add a dimension-level identification-risk annotation layer on top of it.

NLP for climate evidence. Climate-NLP systems verify claims against evidence (CLIMATEFEVER (Diggelmann et al., 2020)), detect environmental claims (Stammbach et al., 2023), analyse corporate climate disclosures with traceable LLM answers (Ni et al., 2023), and tune models for faithful evidence-based question answer-

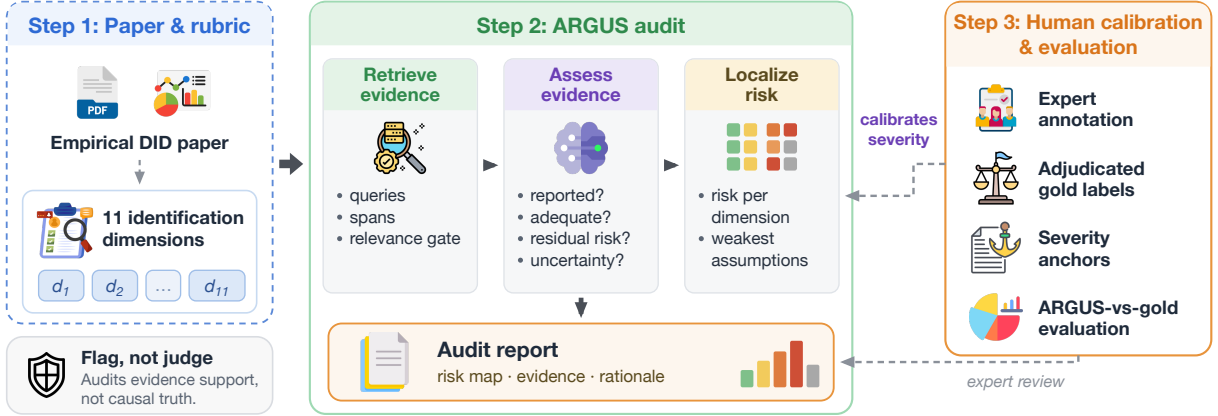


Figure 1: Overview of ARGUS. Step 1 converts an empirical DID paper into an auditable structure using an 11-dimension identification rubric. Step 2 retrieves dimension-specific evidence, assesses reported-evidence adequacy with a bounded LLM audit module, and localizes risk in a human-reviewable report. Step 3 uses expert annotation and adjudicated gold labels to evaluate and calibrate audit severity offline; deployment requires only expert review of the report. ARGUS flags dimensions for expert review.

ing (Schimanski et al., 2024). These systems ground answers in documents; ARGUS extends document-grounded verification from *claims* to the *identification argument* of a causal study, auditing whether reported evidence supports each assumption.

Evaluation methodology and bounded agents.

Our evaluation follows benchmarking lessons: holistic evaluation (Liang et al., 2023) over a single number (we report a layered retrieval/relevance/adequacy/risk profile); dynamic, adversarial benchmarking (Kiela et al., 2021) (which inspires our controlled flaw injection); and the caution that operationalizing a capability is non-neutral (Raji et al., 2021; Gebru et al., 2021) (§6). A closely related concern is that an agent benchmark can measure its harness rather than its model: Zhang et al. (2026b) show that a fixed scaffold making the execution-critical decisions and a scorer grading output shape mask each other, so repairing either alone leaves the ranking uninformative. Our compute-graded ablation (§5.1) is the same move applied inward, separating what the rubric and retrieval layer contribute from what the model does. Where ReAct-style agents (Yao et al., 2023) choose their own action sequence, ARGUS *bounds* the paradigm (agency confined to two scoped sub-tasks behind deterministic control flow), trading autonomy for reproducibility and evidence-traceability.

3 The ARGUS Framework

3.1 Identification rubric

ARGUS decomposes a DID study into eleven auditable dimensions: parallel trends, no anticipation, staggered-timing handling, SUTVA (no interference)/spillovers, control-group construction, specification, inference, sample-period selection, concurrent policies, placebo/robustness, and data measurement. Each dimension is expressed as an *assumption* (what must hold for identification), a testable *implication* (what should therefore be observable), and the *evidence* a credible paper would report to support it. For a carbon-pricing or emissions-trading evaluation, for instance, the parallel-trends dimension expects an event-study or pre-trend test with pre-period estimates near zero (an insignificant pre-trend alone can reflect low power; Roth, 2022), while no-anticipation turns on the gap between a scheme’s *announcement* and *implementation* dates. The rubric is declarative, stored in a configuration file (`identification_dimensions.yaml`), so the audited dimensions are explicit and inspectable rather than buried in code. A companion flaw taxonomy (§4) maps each known identification threat one-to-one onto a dimension, which both documents the rubric’s coverage and supplies the perturbations used for evaluation.

3.2 Bounded LLM pipeline

The audit runs a fixed, deterministic sequence, *decomposition* → *extraction* → *assessment* → *localization* → *report*, shown in the top band of Fig-

ure 2. The stage order is written in code and never chosen by a model. *Decomposition* instantiates the rubric for the paper at hand; *extraction* gathers, per dimension, the in-paper evidence relevant to that assumption; *assessment* evaluates each assumption–implication–evidence chain and emits a per-dimension risk judgement with a rationale and cited evidence; *localization* aggregates these judgements into a risk map ranking where support is weakest; and *report* renders a transparent, evidence-grounded document for a human reviewer.

Agency is confined to the two stages where open-ended judgement actually helps: extraction and assessment. Each is designed as a bounded ReAct-style loop with three constraints: (i) a *small fixed tool set* passed in per call: `evidence_search` over the paper’s text, figures and tables, `figure_parse` for structured figure/table content, and `policy_lookup` for policy metadata such as announcement versus implementation dates; (ii) a *hard step budget* (`max_steps`) so a run cannot wander indefinitely; and (iii) a *logged trace* of every step, tool call, and cited piece of evidence. All other stages are plain orchestration with no agent loop. In the configuration evaluated in §5 the step budget is minimal (`max_steps=1`), and the LLM assessment path resolves to a fixed per-dimension sequence (deterministic lexical retrieval, one relevance-gate call, one adequacy call) that never invokes `figure_parse` or `policy_lookup`; figures and tables enter only through their captions. The gate labels each retrieved passage *high*, *partial*, or *none*: a dimension with no relevant passage is scored *unknown* without an adequacy call, while one with only partially relevant passages is still judged (Appendix D). We therefore call the design *bounded* rather than *agentic*: the evaluated system is structured, dimension-scoped prompting over retrieved evidence with an explicit abstention path, not an open-ended agent, and for an audit we treat this near-determinism as a feature.

Why bound the agency. A fully autonomous agent would make per-run behaviour path-dependent and high-variance, undermining the central evaluation claim of §4: that flaw injection provides *reproducible* local ground truth. Restricting agency to two clearly scoped sub-tasks preserves reproducibility (stable, comparable met-

rics) and evidence-traceability (every judgement anchored to logged evidence a human can inspect). The expert sits at the end of the pipeline: ARGUS surfaces and localizes identification risk; the judgement itself stays with the human.

3.3 Output

ARGUS emits study–dimension–risk triplets, each with an evidence-grounded rationale, alongside the human-readable report. It never produces a verdict on the true causal effect. This output shape keeps the system both auditable by a human and measurable by the evaluation that follows.

4 Evaluation by Flaw Injection

4.1 Local ground truth without the counterfactual

The defining difficulty in evaluating a causal-identification auditor is that the quantity of ultimate interest (the true causal effect of a carbon price, an emissions cap, or an emission standard) is never observed. We therefore do not attempt to certify that ARGUS recovers true effects. Instead we construct *local* ground truth by controlled perturbation. Beginning from a short synthetic fixture, an environmental-policy DID study that reports supporting evidence on all eleven dimensions, we inject exactly one known flaw that targets a single identification dimension, drawn from the flaw taxonomy (`flaw_taxonomy.yaml`; Appendix B traces each flaw to the literature that names it). Because we control which dimension we damaged, we know the correct audit outcome for that dimension, without ever needing the true effect. The bottom band of Figure 2 shows the loop: the clean and injected versions are each audited, and three metrics are computed.

- **Detection:** does the injected version raise risk on the targeted dimension to *medium* or above? (All reported results use this threshold.)
- **False alarm:** does the clean version stay quiet on that dimension?
- **Localization:** is the targeted dimension the one the auditor flags most strongly?

4.2 Realistic perturbation, not sentinels

How a flaw is injected is a methodological choice that matters. A naïve injector appends a sentence

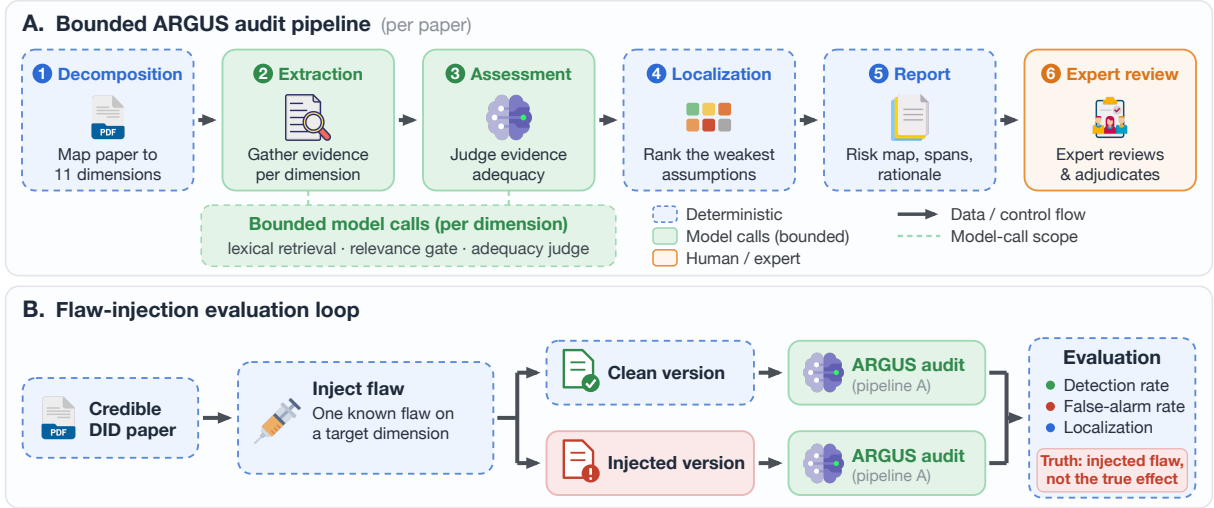


Figure 2: ARGUS architecture. The top band is the audit pipeline, a fixed deterministic sequence; model calls are confined to two stages (extraction and assessment). As evaluated, these resolve per dimension to a fixed call sequence (lexical retrieval, one relevance-gate call, one adequacy call) with a step budget of one and a logged trace; the designed tools `figure_parse` and `policy_lookup` are not exercised (§3.2). The bottom band shows the flaw-injection evaluation loop, which injects one known flaw, audits the clean and injected versions, and scores detection, false alarm, and localization against the injected flaw as local ground truth, never against the unobserved true effect.

that *names* the flaw: e.g. “treated and control have non-zero pre-period coefficients with significant leads.” If the detector then searches for exactly those phrases, detection becomes a tautology: the injector plants the very string the detector matches on. In our initial implementation this produced a perfect-looking but meaningless detection rate of 1.000.

We replace this with *realistic structural perturbation* of two kinds. *Omission* flaws *remove* the section, figure, or table that supplies the support for a dimension (e.g. deleting the placebo-robustness section or the balance table), modelling a paper that simply omits the check; an auditor can flag this non-circularly by noticing the *absence* of evidence. *Commission* flaws *rewrite* a section in flawed-but-natural prose, the way a real flawed paper would describe a mistaken design choice (e.g. pooling all province-years in a two-way fixed-effects regression under staggered policy timing), *without* containing any of the phrases the detector looks for. We enforce the latter with a regression test that fails if an injected paper contains the target dimension’s negative-signal phrases, so the circularity cannot silently return. Detection therefore cannot be achieved by string matching.

5 Results

We evaluate ARGUS along four axes: a synthetic flaw-injection protocol, an architecture ablation, a run over 26 papers tagged difference-in-differences (DID) in an economics corpus that stress-test the audit architecture across empirical economics, beyond the climate-policy setting (carbon pricing, emissions trading, emission standards) that defines its target deployment, and a pilot with adjudicated expert labels. Four findings emerge: (i) LLM pipelines that judge evidence *adequacy* detect planted flaws that a keyword-presence pipeline misses; (ii) per-dimension decomposition drives detection, while the bounded retrieval gate turns retrieval failure into an auditable unknown signal; (iii) on real papers the dominant practical bottleneck shifts from causal reasoning to evidence grounding; and (iv) against a human gold ARGUS is systematically over-severe, a bias that rules derived on the same pilot reduce in-sample.

5.1 Adequacy pipelines versus keyword presence; what the gate trades

We compare three assessors under one flaw-injection protocol on the `clean_supported` fixture (one flaw injected per run, all eleven flaws; `max_steps=1`). A deterministic **keyword baseline** scores each dimension from the presence of

Table 1: Synthetic flaw injection (clean_supported, eleven flaws; all LLM rows gpt-4o, detection replicated on the dated 2024-11-20 snapshot; see Appendix F on serving drift). Every row is an end-to-end pipeline with its own evidence path (keyword and two-stage do not share retrieved evidence, Appendix D; Appendix K holds it fixed); the ablation arms share the model and rubric but remove retrieval, the gate, or both. The final row is the leaked, pre-fix circular setup. On the harder 33-variant benchmark the two-stage assessor rates one clean dimension medium, a 0.09 over-flag (§5.2). Wilson 95% intervals for the main rates are in Table 8.

Assessor	calls	detection	false alarm	localization
keyword baseline	0	0.182	0.000	0.182
single-pass	1	0.818	0.000	0.818
per-dimension, no retrieval	11	1.000	0.000	1.000
two-stage ARGUS	22	0.727	0.000	0.727
(circular, pre-fix)	–	1.000	0.000	1.000

supporting and risk-indicating vocabulary. The **two-stage ARGUS assessor** judges evidence *adequacy* (whether the reported evidence actually establishes the assumption). The two are complete pipelines with their own evidence paths: the baseline scores paragraph chunks, while the two-stage path re-retrieves and gates whole sections (Appendix D). The pair is therefore an *end-to-end* comparison and cannot separate the judgement policy from retrieval. A **single-pass** ablation removes the architecture entirely: the full paper and all eleven dimensions in one gpt-4o call, with the same model, temperature (the sampling-randomness setting, held at 0), risk schema, and scoring harness, but no staged retrieval, relevance gating, or bounded tool use; its strict output schema admits only low, medium, or high, so this arm cannot abstain (Appendix E). Table 1 reports the headline metrics.

The keyword baseline is blind to commission flaws. The baseline never raises a false alarm and correctly flags the two omission flaws (measurement_break, spillover_contamination) by the *absence* of their vocabulary, but beyond those detection is only 0.182 (Figure 3b). Two failure modes explain it: *cross-section vocabulary bleed* (deleting the placebo section does not help when another section also mentions “placebo”) and blindness to *every* commission flaw, whose flawed-but-plausible evidence carries the expected topic vocabulary.

LLM adequacy pipelines close the gap. Replacing the keyword pipeline with the two-stage LLM pipeline raises detection from 0.182 to 0.727 with no false alarms, catching *every* commission flaw the keyword scorer is blind to; handing each dimension the full fixture (the per-dimension arm) reaches 1.000. Since retrieval also differs, a pre-specified control holds the evidence fixed (Appendix K). Reading the keyword pipeline’s own evidence chunks, with no gate, the adequacy judge detects 10 of 11 flaws against the keyword scorer’s 2 (exact McNemar $p=0.008$), and 31 against 8 of the 33 variants. The gain is a trade: the judge rates one clean dimension medium and 11–16% of non-target dimensions on injected papers, where the keyword scorer flags none. The model reads flawed-but-natural prose

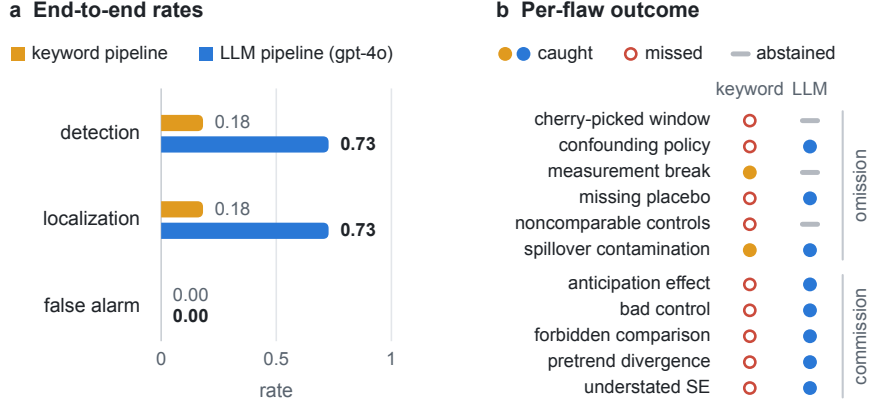


Figure 3: Keyword pipeline vs. LLM evidence-adequacy pipeline ($n=11$ flaws, realistic injection). **(a)** End to end, the LLM pipeline raises detection and localization $0.182 \rightarrow 0.727$ with false alarm 0 (the two pipelines retrieve evidence differently; Appendix K holds it fixed). **(b)** Per-flaw caught/missed: the keyword baseline is *blind to every commission-type flaw*, all of which the LLM catches; the LLM’s three misses are omissions on which it abstains. Table 2 reports the harder 33-variant benchmark.

0.64–0.70, in line with its behaviour on the eleven clear flaws.

Where the gain comes from. A compute-graded ablation on this benchmark (Appendix C) separates the pipeline’s ingredients, de-scaffolding the auditor in the sense of Zhang et al. (2026b): a full-paper single pass (one call) detects 0.727; per-dimension prompting without retrieval (eleven calls) detects 0.879; the full two-stage pipeline (twenty-two calls) detects 0.747. Task *decomposition* drives synthetic detection; the retrieval-and-gate layer then *trades* omission detection for abstention (on these small fixtures a gateless assessor sees deleted sections directly in its context; on real 30-page papers it instead commits judgements nearly everywhere, §5.3). An *oracle-retrieval* arm completes the attribution: fed the target section directly (empty for omissions, bypassing the gate), the ass

Table 2: Harder 33-variant benchmark (gpt-4o, mean of three runs at temperature 0; detection decisions identical on 32/33 variants). Adequacy reasoning catches *commission* flaws; *omission* misses are abstentions to *unknown*, marking the evidence-grounding bottleneck rather than wrong judgements. Intervals in Table 8.

variant type	n	detection	localization
commission	22	0.89	0.82
omission	11	0.45	0.33
all	33	0.75	0.66

same over-severity we quantify against the human gold (§5.5); the clean audit is itself not deterministic. Table 1 shows no false alarm on the original eleven-flaw set.

5.3 From synthetic fixtures to real papers: a relocated bottleneck

We next audit the 26 papers tagged difference-in-differences in the CausalVerify corpus (Zhang et al., 2026a) (Appendix H gives its composition and the limits of that tag). There is no injected ground truth; we report the raw per-dimension risk distribution of each assessor (Figure 5). The keyword baseline is permissive: 91% of its judgements are *low*. It sees topic vocabulary, declares the dimension supported, and flags almost nothing. The two-stage LLM is differentiated, and its most informative output is that $\sim 40\%$ of judgements are *unknown*: the relevance gate could not surface evidence, so the system says so rather than over-flagging. Two regimes appear: dimensions whose discussion is diffuse or lives in figures the text retrieval cannot reach are dominated by *unknown*, while dimensions where a section is retrieved are judged inadequate. Moving to real papers is bottlenecked by *evidence grounding* more than by causal reasoning. The design therefore reports retrieval quality and an explicit *unknown* state distinct from a substantive high-risk judgement.

Why fixtures abstain but real papers flag. The synthetic omission misses of §5.2 become abstentions, yet on real papers a missing check usually becomes a substantive *high*. The asymmetry is one branch of the gate. Retrieval is lexical (Appendix D): deleting a section from the short fixture leaves no passage that even shares its vocabulary, the gate finds nothing relevant, and the pipeline abstains without calling the judge. A full paper usu-

ally contains a section that shares the vocabulary, so the gate returns *partial* relevance, the judge is called on an off-target passage, reports the evidence as missing, and the prompt maps missing to *high*. The pilot shows the pattern (Table 7): all 22 failed-retrieval cells are *unknown*, and 20 of the 22 weak-retrieval cells are *high*, 19 of them with `evidence_status=missing`. This is the branch the calibration layer of §5.6 targets.

Judgements cite specific evidence (Appendix I gives a qualitative example). Roughly 44% of judgements on these papers are *high*; without dimension-level expert labels we cannot yet say whether this is warranted or over-strict where retrieval succeeds. We claim the contrast in behaviour and the localization of the grounding bottleneck, not the correctness of any individual flag.

5.4

5.6 Calibration: from diagnosis to improvement

Of ARGUS’s over-severe *high* judgements, 23/24 have `evidence_status=missing` and 20/24 have `retrieval_quality=weak`. ARGUS treats “weak retrieval found nothing” as a substantive *high*, conflating a *retrieval failure* with the paper lacking the evidence, which the human resolves to low/medium: a fixable calibration target. The pattern is not confined to a single paper: it re-emerges in every leave-one-paper-out fold of the pilot (a recurrence check; no rule is refitted on the other four papers). (This analysis uses a re-run of the pipeline that logs the retrieval-quality and evidence-status fields; it differs from the corpus run of §5.5 in one cell, so its pre-calibration figures are 29/33 over-severe and exact agreement 0.12.)

A deterministic rule over fields the LLM already produces reduces the over-severity: a weak-retrieval *high* (20 cells) is demoted to *medium* (keep coverage) or to *unknown* (abstain). Keeping coverage, the rule raises exact agreement from 0.12 to 0.36 and cuts over-severe cells from 29 to 21: it fixes eight cells, breaks none, and keeps the one high gold cell ARGUS had caught. Abstaining only removes those 20 answers and fixes nothing. Three further single-cell rules lift agreement to 0.45; they are crude field tests, listed in Appendix J and reported as exploratory (Table 14, Figure 6). Every rule was written after an error analysis of this same pilot, so the results are in-sample: the lift points the same way in four of five papers (one tie) and is untested on unseen papers (Appendix G). Calibration corrects severity but cannot recover missing evidence; correct *coverage* needs better grounding.

6 Discussion

What ARGUS measures. A measurement is only as good as the link between the construct and the task that operationalizes it (Jacobs and Wallach, 2021; Raji et al., 2021). ARGUS audits whether a climate-policy study’s identification assumptions are *adequately supported by its reported evidence*: reported-evidence adequacy is an operational proxy for identification credibility, not causal validity itself. The two can diverge, so ARGUS surfaces risk for an expert to weigh and makes no ruling on the effect itself.

Why climate policy. The identification challenges above stress several assumptions at once, and climate-policy estimates feed directly into governance and resource allocation, so a confidently wrong credibility judgement is costly; ARGUS therefore abstains when its gate finds no relevant passage.

Evidence grounding is the dominant practical constraint. A further gap separates the construct from the task: retrieving the relevant evidence. Our real-paper runs show it limits coverage. A reasoning assessor handed off-target evidence over-flags (20 of 22 weak-retrieval pilot cells are rated high), so ARGUS reports a *retrieval-quality* status and an explicit *unknown* risk distinct from a substantive high-risk judgement. This is why the pipeline separates retrieval, relevance filtering, and adequacy assessment.

7 Conclusion

The causal effect of a climate policy is unobservable; the *identification credibility* of the study estimating it is a tractable target for AI assistance. ARGUS operationalizes it: a bounded pipeline turns identification into auditable dimensions and evidence-grounded risk reports; flaw injection supplies reproducible local ground truth without the true effect; and a human-gold diagnostic pilot shows expert labels can both evaluate the auditor and yield

flaw to the literature that names it, so the threat catalogue is inherited while the injector’s prose is ours); flaws authored by independent experts are future work. The benchmarks are small: at $n=11$ and $n=33$ no pairwise difference between architecture arms reaches significance under a paired exact test, and the human-gold intervals are wide (Appendix G), so the directions of the reported rates, not their magnitudes, are the claim. Those intervals and the cell-level paired tests also treat the 55 pilot cells as independent although they are nested in five papers, and the calibration rules were derived on the pilot they are scored on, so the calibration lift is an in-sample result whose size on unseen papers is untested. The parallel-trends criterion is likewise a reporting check: ARGUS does not test whether a pre-trend test had the power to detect a violation. The real corpus is a general-economics stress test that contains no climate-policy evaluation in the narrow sense (Appendix H); a climate-policy corpus run is the natural next test. Relatedly, flaw injection certifies the auditor against the *injected* threat on a single dimension only and inherits the taxonomy’s blind spots, so we treat detection and localization as necessary rather than sufficient, pairing them with the false-alarm rate and the human-gold comparison. Evidence grounding limits coverage. ARGUS emits unknown only when the relevance gate finds no suitable passage; partial matches still reach the judge and can produce over-severe verdicts. Finally, ARGUS screens whether reported evidence supports a study’s identification assumptions and does not adjudicate the true causal effect, which is never observed. Its outputs surface risk for expert review and do not replace it.

Acknowledgments

We thank the anonymous reviewers, whose comments prompted the uncertainty quantification in Appendix G, the account of the retrieval mechanism in Appendix D, and the analysis in §5.3 of why the system abstains on short fixtures yet flags missing evidence on full papers. The icons in Figures 1 and 2 are from Flaticon.com.

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A Domain Motivation and Design Rationale

A.1 Why climate-policy DID is a demanding audit setting

Climate-policy evaluations present several recurring identification challenges. First, policy announcement, legal adoption, and actual implementation may occur at different dates, creating scope for anticipation. Second, carbon markets and environmental regulations are commonly introduced at different times across jurisdictions, making heterogeneous and staggered treatment effects central rather than exceptional. Third, emissions, investment, and production responses can spill across regional borders, challenging no-interference assumptions. Fourth, climate policies frequently overlap with industrial, energy, and air-pollution regulations, complicating the exclusion of concurrent interventions. Fifth, policy effects may

emerge with long and heterogeneous lags. Finally, emissions and environmental outcomes are assembled from administrative, modeled, or remotely sensed sources whose measurement properties vary across places and time. These features do not make DID unique to climate policy, but they make the quality of identification evidence especially consequential for policy interpretation and comparison.

A.2 Identification auditing and research reliability

ARGUS addresses one component of a broader research-reliability problem: published conclusions can vary with specification choices, undisclosed analytic flexibility, and researcher degrees of freedom (Leamer, 1983; Ioannidis, 2005; Simmons et al., 2011; Gelman and Loken, 2014; Silberzahn et al., 2018; Brodeur et al., 2020; Botvinik-Nezer et al., 2020). This is the identification-credibility component of the broader

Table 3: Each injected flaw, its target dimension, and the literature that names the threat.

flaw	dimension	threat named in
pretrend divergence	parallel trends	(Roth, 2022; Roth et al., 2023; Rambachan and Roth, 2023)
anticipation effect	no anticipation	(Callaway and Sant’Anna, 2021; Roth et al., 2023)
forbidden comparison	staggered timing	(Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoeuille, 2020; Sun and Abraham, 2021)
spillover contamination	SUTVA / spillovers	(Athey and Imbens, 2017; Roth et al., 2023)
noncomparable controls	control group	(Callaway and Sant’Anna, 2021; Athey and Imbens, 2017)
bad control	specification	(Angrist and Pischke, 2009)
understated SE	inference	(Bertrand et al., 2004)
cherry-picked window	sample period	(Simmons et al., 2011; Brodeur et al., 2020)
confounding policy	concurrent policies	(Meyer, 1995)
missing placebo	placebo / robustness	(Bertrand et al., 2004; Athey and Imbens, 2017)
measurement break	data measurement	(Meyer, 1995)

research-design literature (Table 3). What we authored is the injector’s prose for each threat, which is why we describe synthetic detection as internal consistency under known threats and treat independently authored flaws as future work.

C Ablation Matrix and Cross-Model Panel

Table 4 grades compute on the 33-variant benchmark: task decomposition (one adequacy call per dimension) drives synthetic detection, and the retrieval-and-gate layer converts omission detection into abstention. Table 5 ports the unchanged pipeline across providers: commission detection is uniformly high, while false-alarm and localization profiles vary sharply by model. Cross-provider decoding defaults differ (newer reasoning models reject an explicit temperature), so these are single runs. Table 6 shows where the operating points diverge: how each model treats a deleted section, how often it already rates the unmodified fixture medium or high, and how often its injected-variant verdict matches gpt-4o’s.

On real papers the gate’s behavioural contrast sharpens. This run used a second single-pass implementation, distinct from the strict-schema arm of Tables 1 and 4: its prompt permits unknown with the instruction to “use sparingly” (Appendix E). Over the 25 papers common to both runs (275 cells) it chose unknown once (0.4%; a model-written abstention with a rationale, not a harness default), where the gated pipeline abstains mechanically on 38.9%. The gated pipeline’s unknown cells become low (1.8% → 14.9%) or medium (14.2% → 43.6%) judgements while the

Table 4: Compute-graded ablation on the 33-variant benchmark (gpt-4o). det / fa / loc = detection, false alarm, localization. Calls per paper are the nominal maximum: when the gate

Table 6: Cross-model operating points on the 33-variant benchmark (single runs; gpt-4o run 1). Omission column: verdicts on the eleven deleted-section variants, in the order unknown / high / medium / low. Clean column: commission pairs (of 22) whose *unmodified* fixture is already rated medium or high. Last column: exact agreement with gpt-4o’s injected-variant verdict.

model	omission verdicts	clean $\geq$ med	agree
gpt-4o	6 / 3 / 2 / 0	2 / 22	–
Claude Opus 4.8	3 / 7 / 1 / 0	18 / 22	20 / 33
Gemini 2.5 Flash	4 / 3 / 3 / 1	4 / 22	26 / 33
Llama 3.1 8B (local)	1 / 10 / 0 / 0	6 / 22	19 / 33

Table 7: Retrieval quality versus emitted risk on the 55 pilot-gold cells (rich re-run). Failed retrieval always abstains; weak retrieval almost always becomes a substantive *high*, 19 of those 20 with evidence_status=missing.

retrieval quality	unknown	high	medium	low
failed (22)	22	0	0	0
weak (22)	0	20	2	0
good (11)	0	5	5	1

with the rubric’s assumption, implication, and expected evidence. The keyword baseline instead searches paragraph chunks capped at 900 characters (sentence-split when longer) with a single query string and top $k=5$, scoring positive and negative sentinel phrases. With max_steps=1 there is no iterative exploration, so breadth is the top- k cap and depth is one round. Chunk size, k , and embedding retrieval were not swept; the oracle-retrieval arm of §5.2 stands in for the retrieval upper bound. Table 7 shows how the gate’s three outcomes map to risk on the pilot-gold cells, the mechanism behind the fixture/real-paper asymmetry of §5.3.

E Prompts

All gpt-4o calls run at temperature 0 (other providers: Appendix F); all but the real-paper single-pass variant at the end of this appendix use a strict JSON schema. Angle-bracketed fields are substituted from the rubric entry or the retrieved passages; nothing else varies across dimensions or papers. The per-dimension no-retrieval arm of §5.2 reuses the adequacy prompt below verbatim, passing the whole paper as the evidence items instead of retrieved sections, so that arm differs from the full pipeline only in what the prompt is fed.

Relevance gate (system).

You are screening retrieved passages for a difference-in-differences (DID) identification audit. For ONE identification dimension, decide whether each passage is actually evidence about THAT dimension’s assumption. This is a relevance check, NOT a quality or risk judgement. Mark a passage ‘high’ if it directly concerns the assumption (e.g. an event-st

your judgement rests on. Keep the rationale to one or two sentences.

Adequacy judge (user).

```
Identification dimension: <name>
Assumption: <assumption>
Testable implication: <implication>
Evidence a credible paper would report:
<expected evidence>
```

```
Evidence retrieved from THIS paper:
- [<source>] <passage text>
```

Judge the adequacy of this evidence for the assumption.

When the gate keeps nothing the evidence block reads (no evidence was retrieved for this dimension), but that path is unreachable in the deployed pipeline: a failed gate short-circuits to unknown without a model call (§5.3).

Schema: {risk, evidence_status, rationale, cited_evidence}.

Single-pass ablation, synthetic benchmarks (system and user).

```
You audit the causal-identification
credibility of a
difference-in-differences (DID) study.
You do NOT judge whether the estimated
effect is true. For EACH of the eleven
identification dimensions listed, judge
whether the evidence the paper reports
is ADEQUATE to support that assumption.
Evidence presence is not evidence
adequacy. Return one judgement per
dimension.
```

```
Identification dimensions to judge:
- <dimension id>: <assumption> (expect:
<expected evidence>)
[... eleven lines ...]
```

```
THE PAPER:
[<source>]
<full section text>
```

Judge the adequacy of the reported evidence for EVERY dimension.

Schema: a judgements array of {dimension, risk, evidence_status, rationale}, with dimension restricted to the eleven rubric ids and risk to low | medium | high; this arm cannot answer unknown. It produced the single-pass rows of Tables 1 and 4.

Single-pass variant, real-paper run (system, abridged to the risk block).

For each dimension output one risk level:

```
low : adequate, credible evidence is reported.
```

```
medium : evidence is partial, indirect, or weak.
```

```
high : the required evidence is absent, or present but flawed.
```

```
unknown : the text gives no basis to judge either way (use sparingly).
```

Also output evidence_status in {sufficient, partial, missing, flawed} (roughly sufficient->low, partial->medium, missing/flawed->high). Be conservative: absent, thin, or flawed evidence is not 'supported'.

The real-paper contrast of Appendix C used this second implementation (single_pass

mark was run three times end-to-end (roughly 750 model calls per run, ~20 minutes and ~US\$8 per gpt-4o run; the local Llama run took ~67 minutes on a laptop at zero API cost); binary detection agreed on 32/33 variants across the three runs and target-dimension risk labels on 31/33, while the clean audit flagged a different dimension in run 1 than in runs 2 and 3, and Table 2 reports the three-run mean. The two-call-per-dimension pipeline audits one paper for roughly US\$0.25 of gpt-4o usage.

G Uncertainty Quantification

Table 8 gives Wilson 95% intervals for the main proportions reported in the main text and exact tests between arms evaluated on the same items; Table 9 gives cell-bootstrap intervals (5,000 resamples) for the human-gold agreement metrics. At $n=11$ the keyword-versus-two-stage gap is marginal under the paired test on the same flaws (7 vs 1 discordant, $p=0.07$; an unpaired Fisher test, which ignores the pairing, gives $p=0.03$); at $n=33$ no pairwise difference between the single-pass, per-dimension, two-stage, and oracle arms reaches significance. For the calibration lift the comparison is paired, so overlap of the intervals in Table 9 is not the test. On the 33 answered cells the weak-retrieval rule alone turns eight wrong cells exact and none the other way (exact McNemar $p=0.008$); all four rules turn eleven ($p=0.001$). Those tests treat cells as independent, yet the cells are nested in five papers. By paper, the single rule improves exact agreement in four papers and ties in one (sign test with the tie dropped, $p=0.125$); four rules improve all five ($p=0.0625$, the smallest value five papers allow). Neither analysis removes the bias of deriving the rules on the pilot they are scored on. We therefore claim a consistent in-sample direction, not a demonstrated out-of-sample gain.

Tables 8 and 9 are computed from committed files by `uncertainty.py` and `calibration_recheck.py` in `experiments/ablations/`; the pilot cells and adjudicated labels they read are frozen with checksums in `experiments/annotation/pilot_frozen/`.

H The Real-Paper Corpus

The 26 papers audited in §5.3 are those tagged difference-in-differences in the corpus of 259 eco-

Table 8: Wilson 95% intervals for reported rates (top) and exact tests between arms (bottom; McNemar tests are paired on the same flaws or variants; discordant = detected by the first arm only / by the second only). False-alarm rows give counts only: every variant is paired with the same clean fixture, so they are repeated readings of one clean audit of eleven dimensions, not independent trials, and an interval would overstate what they show. Detection intervals describe this fixed test set (33 variants nested in eleven dimensions of one fixture), not real papers at large.

benchmark	quantity	k/n	95% CI
11 flaws	keyword detection	2/11	0.05–0.48
	single-pass detection	9/11	0.52–0.95
	per-dimension detection	11/11	0.74–1.00

Table 9: Cell-bootstrap 95% intervals for agreement with the adjudicated gold (answered cells only; quadratic-weighted κ ; 5,000 resamples). Rule 1 is the weak-retrieval rule; rules 1–4 add the three single-cell rules of Appendix J. Resampling cells ignores their nesting in five papers, so the intervals are too narrow.

policy	n	exact agreement	weighted κ
before	33	0.12 (0.03–0.24)	0.06 (–0.02–0.15)
rule 1, demote→med	33	0.36 (0.21–0.55)	0.21 (0.02–0.39)
rule 1, abstain→unk	13	0.31 (0.08–0.54)	0.32 (0.04–0.58)
rules 1–4, demote→med	33	0.45 (0.30–0.61)	0.13 (–0.07–0.37)
rules 1–4, abstain→unk	13	0.54 (0.31–0.77)	0.25 (–0.17–0.69)

Table 10: Venue of the 26 audited papers, as recorded for the version held in the source corpus. Two entries carry a corrupted venue field in the corpus index; they are counted under the journal that published them (Journal of Political Economy, 2003; Review of Economic Studies, 2002).

venue / version	papers
American Economic Review	4
Quarterly Journal of Economics	3
Journal of Political Economy	2
Journal of Financial Economics	2
AEJ: Applied Economics	2
JF / REStat / REStud / J. Monetary Econ. / J. Int. Econ. (one each)	5
New England Journal of Medicine	1
NBER working-paper versions	7

(2), and industrial location (1); the environment-and-energy subset covers the industrial-activity effects of environmental regulation, air quality and early-life mortality, nonlinear electricity pricing, and gasoline prices and vehicle choice. None of the 26 is a climate-policy evaluation in the narrow sense (a carbon price, an emissions trading scheme, or an emission standard): the real corpus is a general-economics stress test of the audit architecture, and the climate-specific content of the paper is the rubric instantiation of §3 and Appendix A; the synthetic fixture of §4 is a generic environmental-policy study (a China emissions-trading fixture ships with the code but produced none of the numbers reported here). Audit behaviour is not driven by domain: the environment-and-energy subset’s risk distribution (high 0.34, unknown 0.43; 44 cells) is close to the remainder’s (high 0.46, unknown 0.39; 242 cells).

I Qualitative Audit Examples

A complete audit report. Table 12 shows ARGUS’s actual output on the synthetic fixture with the *anticipation* flaw injected (§4): the report flags exactly the damaged dimension as high, giving the anticipation-gap rationale, keeps nine supported

Table 12: ARGUS audit report (abridged) on the fixture with the injected anticipation flaw. Rationales are abridged from the evidence-cited report; the injected dimension is the one flagged high.

dimension	risk	rationale (abridged)
no anticipation	high	outcomes began changing between the policy announcement and its rollout, undermining no-anticipation
inference	medium	clustering and wild-cluster bootstrap reported, but the number of clusters is not stated
parallel trends	low	event-study with null pre-period coefficients and placebo tests
treatment timing	low	heterogeneity-robust estimators with a Goodman-Bacon decomposition
7 further dimensions	low	supporting evidence retrieved and judged adequate

0.17) but almost entirely ordinal (weighted $\kappa = 0.31$; 96% of disagreements are a single severity step). Of the 27 round-1 disagreements, 16 resolved to a single label once the low/medium boundary was anchored, and 11 were flagged intrinsically ambiguous. In the leave-one-paper-out check (§5.6), the demotion rule’s trigger conditions recur in all five folds: among each fold’s over-severe *high* cells, 79–86% have weak retrieval and 94–100% have missing evidence (computed on the rich re-run of §5.6). The folds test whether the error pattern recurs; no rule was refitted on four papers and scored on the fifth.

The calibration rules. The layer acts on *high* cells only and never raises a risk; the first matching rule applies. Rule 1 (20 cells): `retrieval_quality=weak` $\rightarrow$ *medium*, or $\rightarrow$ *unknown* under the abstain policy. Three further rules each fire on one cell and demote to *medium*: rule 2, the dimension is staggered timing and the rationale names a modern estimator (Callaway, Sun–Abraham, Goodman-Bacon); rule 3, the dimension is inference, `evidence_status=missing`, and the rationale mentions “cluster” or “robust standard errors”; rule 4, the dimension is concurrent policies and `retrieval_quality=good`. Rules 2–4 are string and field tests written minutes after the over-severity analysis of this pilot, and all three cells they touch land on the gold label. They are not principled: a paper’s age does not remove a staggered-timing bias, a rationale that mentions clustering may be reporting its absence, and rule 4 has no content condition at all (locating relevant evidence is not the evidence supporting the assumption). We treat only rule 1 as a result. The released script (`calibrate_argus_`

Table 13: ARGUS-LLM vs adjudicated human gold (55 cells, 5 papers) under two *unknown* policies: coverage (drop abstentions) and strict (*unknown* counts as mismatch). Pilot-scale: only 2 gold cells are *high*; the robust signal is the one-directional over-severity and the abstention rate.

unknown policy	<i>n</i>	exact	κ	wt. κ
exclude (coverage)	33	0.152	0.019	0.044
mismatch (strict)	55	0.091	0.002	–

Table 14: Deterministic calibration vs

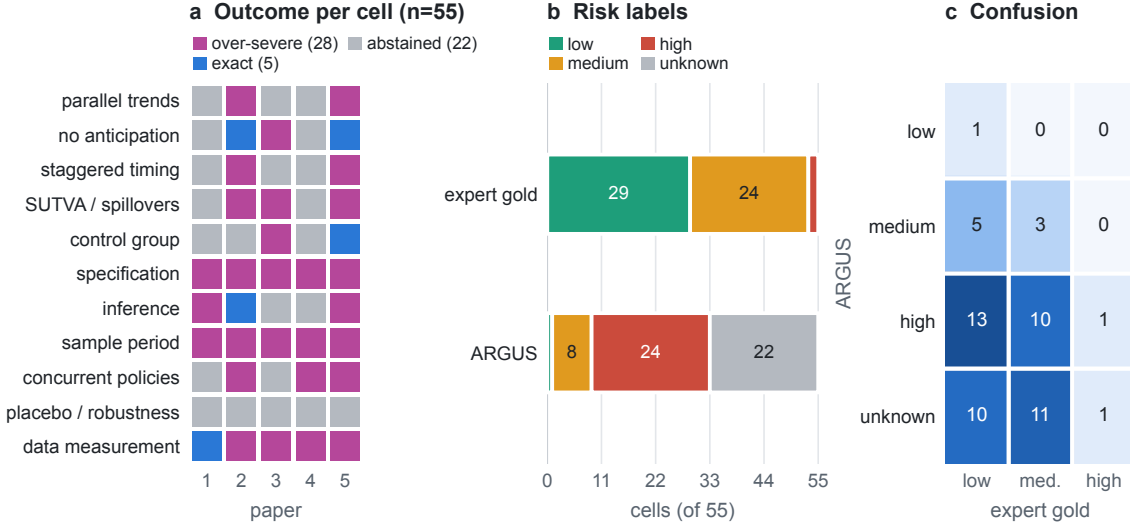


Figure 4: ARGUS-LLM versus the adjudicated expert gold on the 55 pilot cells (frozen protocol). **(a)** Every cell, eleven dimensions by five papers, coloured by outcome: ARGUS abstains on 22 and, of the 33 it answers, is more severe than the gold on 28, matches it on 5, and is less severe on none. **(b)** Distribution of risk labels in the gold and in ARGUS’s output. **(c)** ARGUS-by-gold confusion matrix (counts).

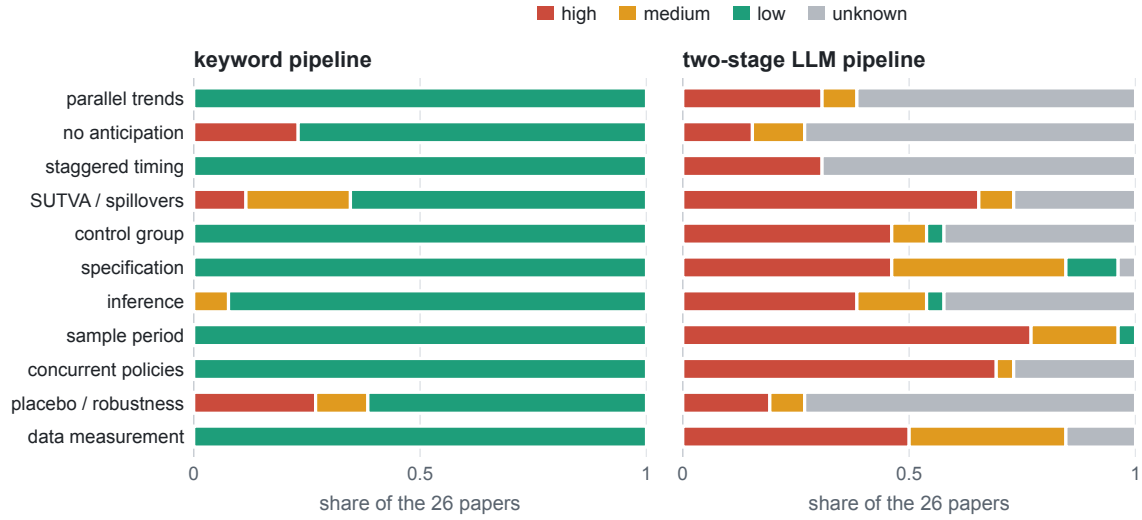


Figure 5: Per-dimension risk distribution on the 26 papers tagged DID in the source corpus. The keyword baseline (left) is almost entirely *low*; the two-stage LLM (right) is differentiated, with a large *unknown* share that marks retrieval failure rather than over-flagging.

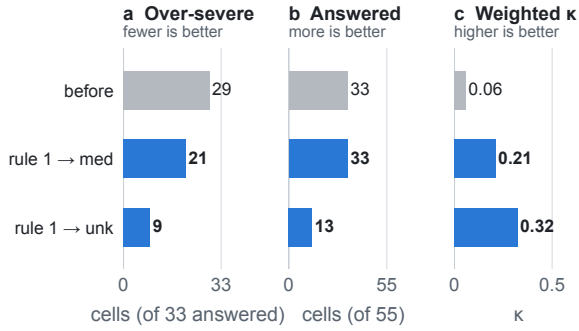


Figure 6: **Calibration trade-off: severity reduction versus coverage** against the adjudicated human gold, for the weak-retrieval rule alone (in-sample; med and unk are the two policies of Table 14, which also gives the exploratory four-rule set). Demoting weak-retrieval *high*s to *medium* preserves coverage (33 answered) while cutting over-severe judgments from 29 to 21; abstaining leaves fewer over-severe cells (9) and a higher weighted κ (0.32) only by discarding 20 answers, and fixes none of them. Calibration reduces severity bias; the remaining bottleneck is evidence grounding, not severity thresholding alone.